4

The Normal Distribution and Standard Scores

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Key Concepts

1. Normal distribution (family of distributions, unit normal distribution, area under the curve, points of inflection, asymptotic curve)
2. Standard scores (z, T, IQ)
3. Symmetry
4. Skewness (positively skewed, negatively skewed)

5. Kurtosis (leptokurtic, platykurtic, mesokurtic)
6. Moments around the mean

In Chapter 3, we continued our discussion of descriptive statistics, which were defined as techniques that allow us to tabulate, summarize, and depict a collection of data in an abbreviated fashion. We considered the following three topics: summation notation (method for summing a set of scores), measures of central tendency (measures for boiling down a set of scores into a single value used to represent the data), and measures of dispersion (measures dealing with the extent to which a collection of scores vary).

In this chapter, we delve more into the field of descriptive statistics in terms of three additional topics. First, we consider the most commonly used distributional shape, the normal distribution. Although in this chapter we discuss the major characteristics of the normal distribution and how it is used descriptively, in later chapters we see how the normal distribution is used inferentially as an assumption for certain statistical tests. Second, several types of standard scores are considered. To this point we have looked at raw scores and deviation scores. Here we consider scores that are often easier to interpret, known as *standard scores*. Third, we examine two other measures useful for describing a collection of data, namely skewness and kurtosis. As we show shortly, *skewness* refers to the lack of symmetry of a distribution of scores and kurtosis refers to the *peakedness* of a distribution of scores. Finally, we provide a template for writing research questions, develop an APA-style paragraph of results for an example dataset, and also illustrate the use of SPSS and **R**. Concepts to be discussed include the normal distribution (i.e., family of distributions, unit normal distribution, area under the curve, points of inflection, asymptotic curve), standard scores (e.g., *z*, *T*, IQ), symmetry, skewness (positively skewed, negatively skewed), kurtosis (leptokurtic, platykurtic, mesokurtic), and moments around the mean. Our objectives are that by the end of this chapter, you will be able to (a) understand the normal distribution and utilize the normal table; (b) determine and interpret different types of standard scores, particularly *z* scores; and (c) understand and interpret skewness and kurtosis statistics.

4.1 The Normal Distribution and How It Works

You may remember the following research scenario that was first introduced in Chapter 2. We will revisit our talented group of graduate students in this chapter as they continue to explore the data.

The graduate students in the statistics lab, Addie Venture, Oso Wyse, Challie Lenge, and Ott Lier, have been assigned their first task as research assistants. Dr. Debhard, a statistics professor, has given the group of students quiz data collected from 25 students enrolled in an introductory statistics course and has asked the group to summarize the data. Working now with Challie Lenge, Dr. Debhard has asked Challie to revisit the following research question related to distributional shape: *What is the distributional shape of the statistics quiz score?* Additionally, Dr. Debhard has asked Challie to standardize the quiz score and compare student 1 to student 3 relative to the mean. The corresponding research question that Challie is provided for this analysis is as follows: *In standard deviation units, what is the relative standing to the mean of student 1 compared to student 3?*

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variable, X or z, is irrelevant in terms of determining such areas. In the next section we deal more with such transformations.

4.2 Standard Scores and How They Work

We have already devoted considerable attention to z scores, which are one type of standard score. In this section we describe an application of z scores leading up to a discussion of other types of standard scores. As we show, the major purpose of standard scores is to place scores on the same standard scale so that comparisons can be made across individuals and/or variables. Without some standard scale, comparisons across individuals and/or variables would be difficult to make. Examples are coming right up.

4.2.1 *z* Scores

You have just interviewed for your dream job. As part of your interview, you completed a cognitive ability assessment (which measured problem-solving skills and ability to learn and understand instructions) and a motivation index (designed to measure work engagement motivation). On the cognitive ability assessment, you receives a score of 75 and on the motivation index you receive a score of 60. The natural question to ask is, "Which performance was the stronger one?" The suspense is killing you! No information about any of the following is available: maximum score possible, mean of the candidates who were interviewed (or any other central tendency measure), or standard deviation of the candidates who were interviewed (or any other dispersion measure). It is possible, and quite likely, that the two assessments had a different number of possible points, different means, and/or different standard deviations. How can we possibly answer our question?

The answer, of course, is to use z scores if the data are assumed to be normally distributed, once the relevant information is obtained. Let us take a minor digression before we return to answer our question in more detail. Recall the formula for standardizing variable X into a z score:

$$z_i = \frac{\left(X_i - \mu_X\right)}{\sigma_X}$$

where the X subscript has been added to the mean and standard deviation for purposes of clarifying which variable is being considered. If variable X is the number of items correct on a test, then the numerator is the deviation of the student's raw score from the class mean (i.e., the numerator is a deviation score as previously defined in Chapter 3), measured in terms of items correct, and the denominator is the standard deviation of the class, measured in terms of items correct. Because both the numerator and denominator are measured in terms of items correct, the resultant z score is measured in terms of no units (as the units of the numerator and denominator essentially cancel out). *Given that z scores have no units (i.e., the z score is interpreted as the number of standard deviation units above or below the mean), this allows us to compare two different raw score variables with different scales, means, and/or standard*

Consider a realistic example where we have a raw score variable we want to transform into a standard score, and we want to control the mean and standard deviation. For example, we have statistics midterm raw scores with 225 points possible. We want to develop a standard score with a mean of 50 and a standard deviation of 5. We also have scores on other variables that are on different scales with different means and different standard deviations (e.g., statistics final exam scores worth 175 points, a set of 20 lab assignments worth a total of 200 points, a statistics performance assessment worth 100 points). We can standardize each of those variables by placing them on the same scale with the same mean and same standard deviation, thereby allowing comparisons across variables. This is precisely the rationale used by testing companies and researchers when they develop standard scores. In short, from z scores we can develop a T, IQ, "stat," or any other type of standard score. Examples of types of standard scores are summarized in Box 4.1.

BOX 4.1 Examples of Types of Standard Scores

Standard Score	Distribution*
Z (unit normal)	$N(0,1)$
College Entrance Examination Board (CEEB) score	$N(500,10{,}000)$
T score	$N(50,100)$
Weschler intelligence scale	$N(100,225)$
Stanford-Binet intelligence scale	$N(100,256)$

*$N\left(\mu,\sigma^2\right)$

4.3 Skewness and Kurtosis Statistics

In previous chapters we discussed the distributional concepts of symmetry, skewness, central tendency, and dispersion. In this section we more closely define symmetry as well as the statistics commonly used to measure skewness and kurtosis.

4.3.1 Symmetry

Conceptually, we define a distribution as being **symmetric** *if when we divide the distribution precisely in one-half, the left-hand half is a mirror image of the right-hand half*. That is, the distribution above the mean is a mirror image of the distribution below the mean. To put it another way, a distribution is **symmetric around the mean** if for every score that is q units below the mean, there is a corresponding score that is q units above the mean.

Two examples of symmetric distributions are shown in Figure 4.4. In Figure 4.4a, we have a normal distribution, which is clearly symmetric around the mean. In Figure 4.4b, we have a symmetric distribution that is bimodal, unlike the previous example. From these and other numerous examples, we can make the following two conclusions. First, if a distribution is *symmetric*, then the mean is equal to the median. Second, if a distribution is

Property	Characteristics	Conventions
	Platykurtic distributions have negative kurtosis values (being flat rather than peaked). Kurtosis values can range from negative to positive infinity. Kurtosis can be computed on variables that are interval or ratio in scale.	*Conservative convention:* kurtosis within ±1.0 are normal. Kurtosis values outside the range of ±2 standard errors of kurtosis suggest a distribution that is nonnormal.

4.4 Computing Graphs and Standard Scores Using SPSS

Here we review what SPSS has to offer for examining distributional shape and computing standard scores. The following tools have proven to be quite useful for these purposes: Explore, Descriptives, Frequencies, Graphs, and Transform.

4.4.1 Explore

Step 1. Explore can be invoked by clicking "Analyze" in the top pulldown menu, then "Descriptive Statistics," and then "Explore." Following the screenshot for "EXPLORE: Step 1" in Figure 4.7 produces the Explore dialog box. For brevity, we have not reproduced this initial screenshot when we discuss the Descriptives and Frequencies tools; however, you see here where they can be found from the pulldown menus.

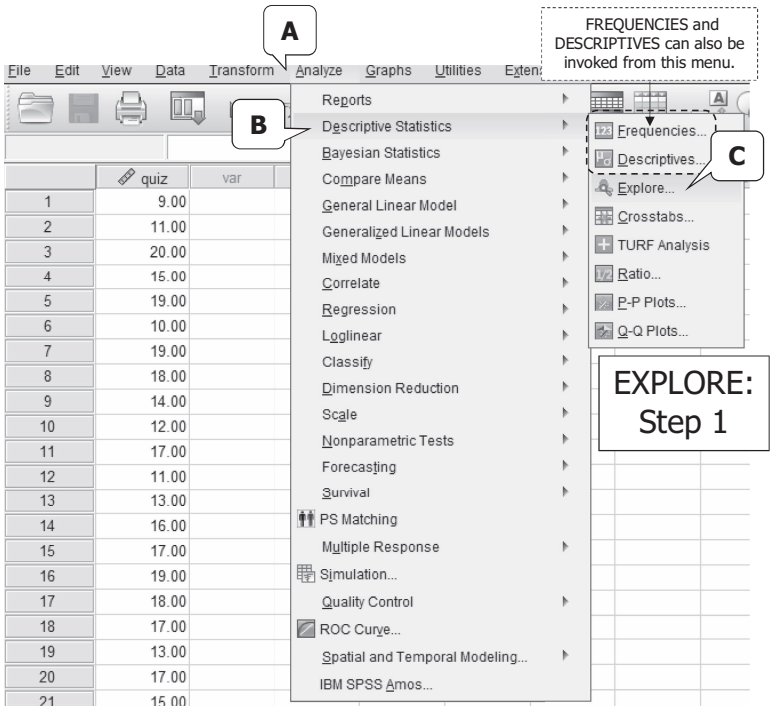


FIGURE 4.7
EXPLORE: Step 1.

4.5 Computing Graphs and Standard Scores Using R

Next we consider **R** for various statistics and graphs. The scripts are provided within the blocks with additional annotation to assist in understanding how the commands work. Should you want to write reminder notes and annotation to yourself as you write the commands in **R** (and we highly encourage doing so), remember that any text that follows a hashtag (i.e., #) is annotation only and not part of the **R** code. Thus, you can write annotations directly into **R** with hashtags. We encourage this practice so that when you call up the commands in the future, you'll understand what the various lines of code are doing. You may think you'll remember what you did. However, trust us. There is a good chance that you won't. Thus, consider it best practice when using **R** to annotate heavily!

4.5.1 Reading Data into R

We will first read in our data (Figure 4.19). We will be working with the Ch4_quiz.csv data.

```
getwd()
```

R is always pointed to a directory on your computer. To find out which directory it is pointed to, run this "get working directory" command. We will assume that we need to change the working directory, and will use the next line of code to set the working directory to the desired path.

```
setwd("E:/Folder")
```

To set the working directory, use the *setwd* function and change what is in quotation marks here to your file location. Also, if you are copying the directory name, it will copy in slashes. You will need to change the slash (i.e., \) to forward slash (i.e., /). Note that you need your destination name within quotation marks in the parentheses.

```
Ch4_quiz <- read.csv("Ch4_quiz.csv")
```

The *read.csv* function reads your data into **R**. What's to the left of the <- will be what you want to call the data in **R**. In this example, we're calling the **R** dataframe "Ch4_quiz." What's to the right of the <- tells **R** to find this particular csv file. In this example, our file is called "Ch4_quiz.csv." Make sure the extension (i.e., .csv) is included in your script. Also note that you need the name of your file in quotation marks within the parentheses.

```
names(Ch4_quiz)
```

The *names* function will produce a list of variable names for each dataframe as follows. This is a good check to make sure your data has been read in correctly.

```
[1] "quiz"
```

```
View(Ch4_quiz)
```

The *View* function will let you view the dataset in spreadsheet format in RStudio.

```
summary(Ch4_quiz)
```

FIGURE 4.19
Reading data into **R**.

```
View(Ch4_quiz)
```

We use the *View* function to confirm that our new variable has been created and added to our dataframe by displaying the dataframe in RStudio.

```
summary(Ch4_quiz)
```

We use the *summary* function to generate basic descriptive statistics on our variable. We see that the mean of our new standardized variable, "Zquiz.v1," is 0, which is expected!

```
quiz                Zquiz.V1
Min.   : 9.00   Min.   :-2.0737630
1st Qu.:13.00   1st Qu.:-0.8092734
Median :17.00   Median : 0.4552163
Mean   :15.56   Mean   : 0.0000000
3rd Qu.:18.00   3rd Qu.: 0.7713387
Max.   :20.00   Max.   : 1.4035835
```

FIGURE 4.22 (continued)
Creating a standardized variable.

4.6 Research Question Template and Example Write-Up

As stated in the previous chapter, depending on the purpose of your research study, you may or may not write a research question that corresponds to your descriptive statistics. If the end result of your research paper is to present results from inferential statistics, it may be that your research questions correspond only to those inferential questions, and thus no question is presented to represent the descriptive statistics. That is quite common. On the other hand, if the ultimate purpose of your research study is purely descriptive in nature, then writing one or more research questions that correspond to the descriptive statistics is not only entirely appropriate, but (in most cases) absolutely necessary.

It is time again to revisit our graduate research assistant, Challie Lenge, who was reintroduced at the beginning of the chapter. As a reminder, Challie was working with Dr. Debhard, a statistics professor. Challie's task was to continue to summarize data from 25 students enrolled in a statistics course, this time paying particular attention to distributional shape and standardization. The questions posed this time by Dr. Debhard were as follows: *What is the distributional shape of the statistics quiz score? In standard deviation units, what is the relative standing to the mean of student 1 compared to student 3?* The following is a template for writing a descriptive research question for summarizing distributional shape (this may sound familiar as this was first presented in Chapter 2 when we initially discussed distributional shape). This is followed by a template for writing a research question related to standardization.

What is the distributional shape of the [variable]? In standard deviation units, what is the relative standing to the mean of [unit 1] compared to [unit 3]?

Next, we present an APA-style paragraph summarizing the results of the statistics quiz data example answering the questions posed to Marie.

The skewness value is −.598 (*SE* = .464) and the kurtosis value is −.741 (*SE* = .902). Skewness and kurtosis values within the range of ±2 (*SE*) are generally considered normal. Given our values, skewness is within the range of −.928 to +.928 and kurtosis is within the range of −1.804 and +1.804, and these would be considered normal. Another convention is that the skewness and kurtosis values should fall within an absolute value of 2.0 to be considered normal. Applying this rule, normality is still evident. The histogram with a normal curve overlay is depicted in Figure 4.15. Taken with the skewness and kurtosis statistics, these results indicate that the quiz scores are reasonably normally distributed. There is a slight negative skew such that there are more scores at the high end of the distribution than a typical normal distribution. There is also a slight negative kurtosis indicating that the distribution is slightly flatter than a normal distribution, with a few more extreme scores at the low end of the distribution. Again, however, the values are within the range of what is considered a reasonable approximation to the normal curve.

Prior to standardization, student 1 had a score of 9 and student 3 had a score of 20. The quiz score data were standardized using the unit normal formula. After standardization, student 1's score was −2.07 and student 3's score was 1.40. This suggests that student 1 was slightly more than two standard deviation units below the mean on the statistics quiz score while student 3 was nearly 1.5 standard deviation units above the mean.

4.7 Additional Resources

In the previous chapters, we have mentioned a number of excellent resources for learning statistics. As we are still in the early stages of learning statistics, there are no additional resources that we suggest that are specifically related to normal distributions and standard scores. Rather, we refer you back to earlier chapters for supplemental resources for statistics as well as statistical software.

Problems

Conceptual Problems

1. For which of the following distributions will the skewness value be zero?
   a. *N*(0,1)
   b. *N*(0,2)
   c. *N*(10,50)
   d. All of the above

2. For which of the following distributions will the kurtosis value be zero?
   a. *N*(0,1)
   b. *N*(0,2)
   c. *N*(10,50)
   d. All of the above